

Chapter 18.A

Chemical Thermodynamics

Chemical Principles I

CHEM 1211K

1. State the second law of thermodynamics.
2. Define entropy and identify the sign of ΔS from chemical equations or changes of state.
3. State and apply the statistical definition of entropy.
4. State the third law of thermodynamics.
5. Compare and contrast ΔS_{univ} , ΔS_{sys} , and ΔS_{surr} .
6. Define Gibbs free energy and describe how the signs of ΔH and ΔS affect the sign of ΔG .
7. Relate the sign of ΔG to the favorability of a reaction.
8. Determine the temperature at which reactants and products are at equal free energy for a given reaction.

- Much of the work of chemists is to make predictions and answer questions about physical and chemical processes.
 - *Reactions and Stoichiometry*: What happens and in what proportions?
 - *Kinetics*: At what speed does a reaction occur?
 - ***Equilibrium and Thermodynamics***: To what extent does a reaction occur, given infinite time?
- The basic question of this chapter is whether a physical or chemical process will or will not occur spontaneously starting from a standard set of initial conditions.

18.1 Spontaneous Processes



- A **spontaneous process** occurs without continuing outside influence—it “just goes.”
- Spontaneity and time are theoretically *unrelated*. Spontaneous processes can be slow!

- Ideally, we'd like a state function that tells us whether a given process is spontaneous or not based on the sign of the state function (or a change therein). Consider gravitational potential energy as a metaphor from physics.
- **Entropy (S)** is a measure of disorder, randomness, or (best) [matter and energy dispersal](#) in a system. It is the state function we need! Consider popping a balloon...
- **The second law of thermodynamics:** spontaneous processes are associated with an increase (or no change) in the entropy of the universe S_{univ} .

18.1 The Classical Definition of Entropy

- *Not tested directly* but critical for conceptual understanding of entropy...
- Classical thermodynamics defines infinitesimal entropy change dS as the infinitesimal heat transferred to or from the system reversibly (δq_{rev}) divided by the temperature T at which this heat transfer occurs:

$$dS = \delta q_{rev}/T$$

- The units of entropy are thus **energy per temperature**, most commonly Joules per Kelvin.
- **Add heat** and the entropy of the system **increases**; **remove heat** and the entropy of the system **decreases**.
- Conceptually, entropy is a measure of the *amount of dispersed energy in the system* divided by the *temperature of the system*. Further reading:
<https://franklambert.net/entropysite.com/#summary>.

18.1 Factors Affecting Entropy



- Factors affecting the entropy of a system (S) or entropy change of a process (ΔS)...
 - Phase of matter
 - Temperature
 - Volume of a gas
 - Amount (in moles) of a gas
 - Amount (in moles) of distinct types of compounds

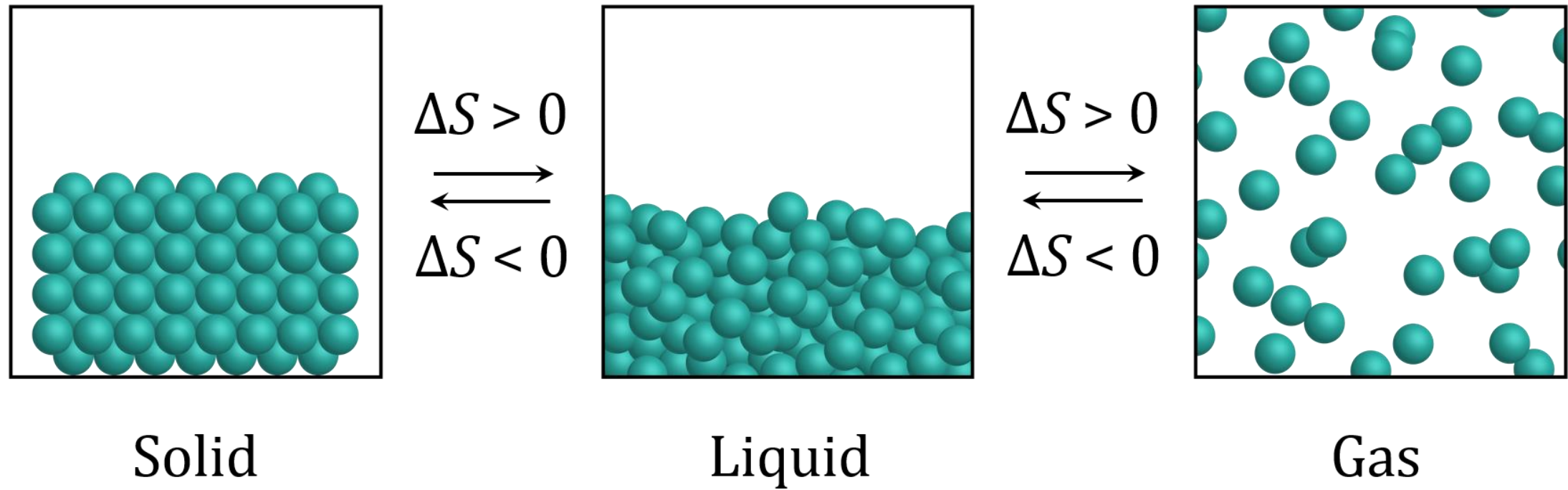
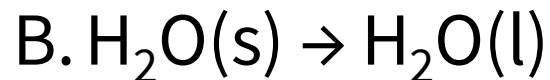
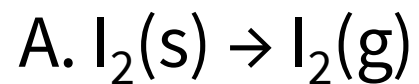


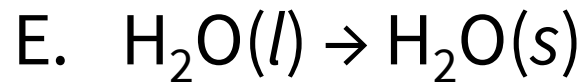
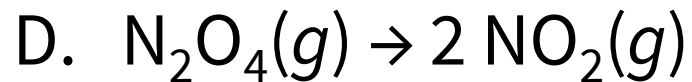
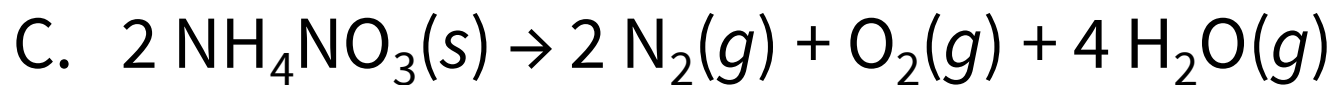
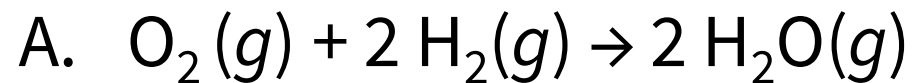
Figure 18.1. Entropy changes for phase transitions involving solid, liquid, and gas.

For which process(es) would you expect the change in entropy to be less than zero? Select all that apply.



E. Condensation

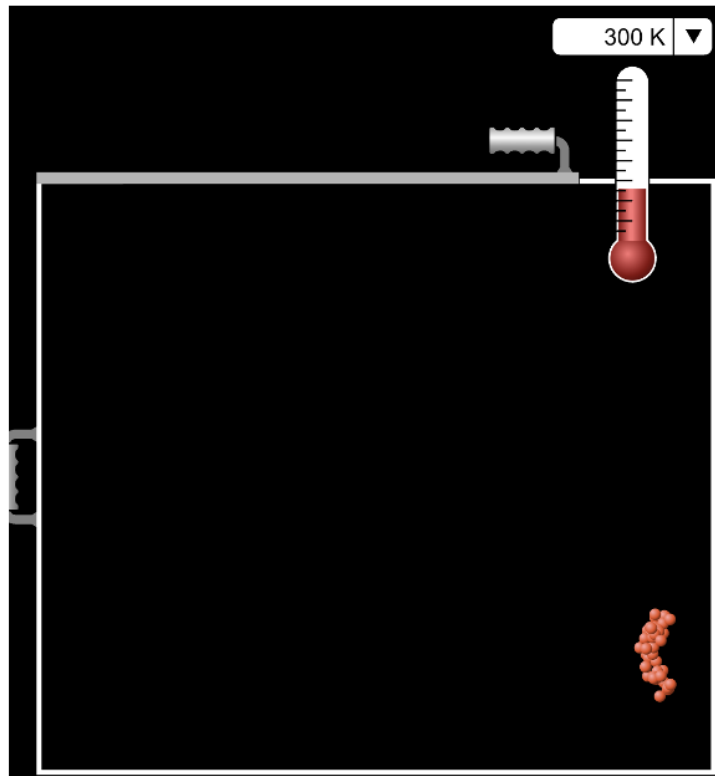
For which process would you expect ΔS° to be *most* positive?



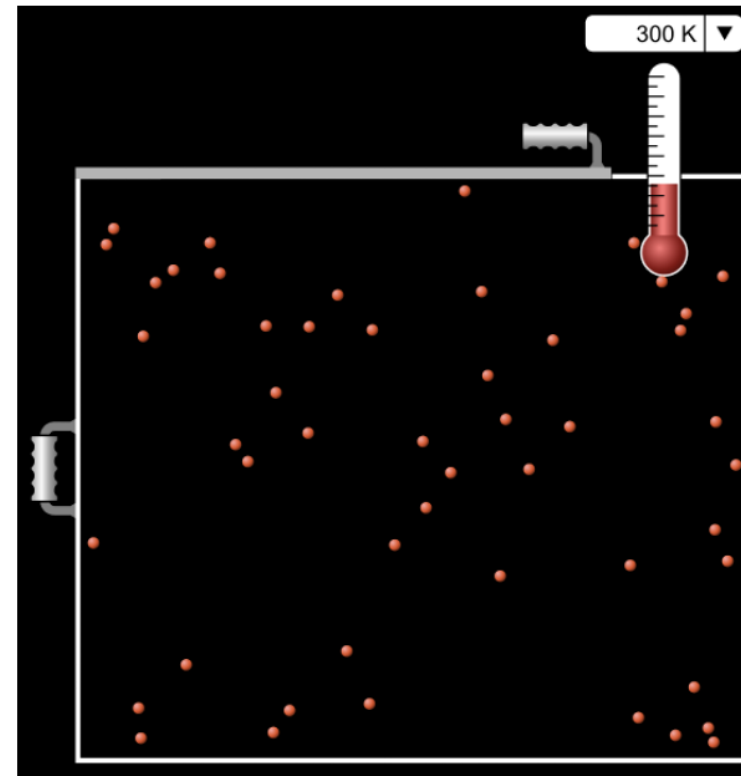
18.1 The Statistical Definition of Entropy | 1 of 4

- Boltzmann observed that systems tend to move spontaneously toward a state of maximum energy dispersal simply because the maximally dispersed state is more *statistically likely* than more concentrated states.
- Consider the following two “snapshots” of an ideal gas at given T and n .
Which feels more likely to observe? What is the direction of the “arrow of time” here?

state 1

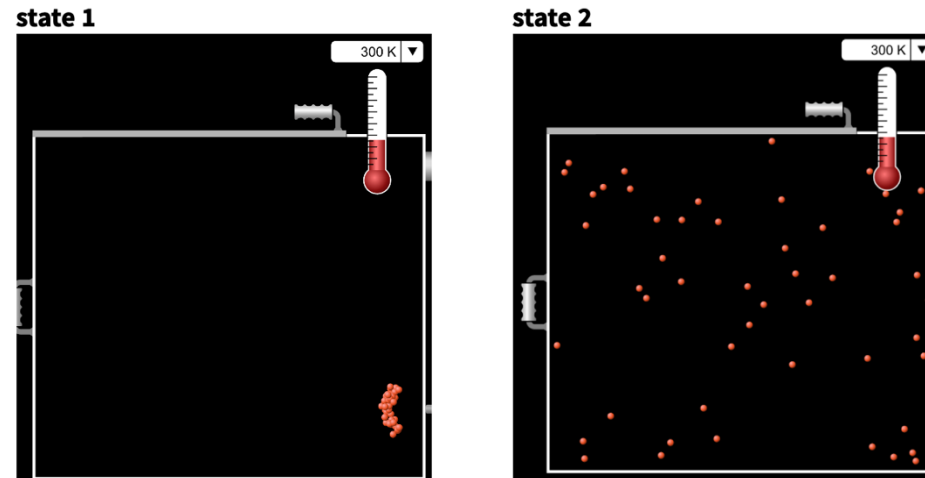


state 2



18.1 The Statistical Definition of Entropy | 2 of 4

- Of course, the arrow of time points from state 1 to state 2. The gas expands spontaneously into the empty space in the container.
 - The effective volume of the gas increases from state 1 to state 2. $\Delta V > 0$.
 - The kinetic energy of the gas particles and the particles themselves disperse throughout the container in the process. Thus, $S_2 - S_1 = \Delta S > 0$.
 - State 1 will _{almost} never arise if the system starts in state 2.
 - The system evolves spontaneously from state 1 to state 2 because state 2 is **exponentially more likely to exist** than state 1.



- The **likelihood** W of a **macrostate** is equal to the number of **microstates** that give rise to that macrostate.
 - The **macrostate** of a system is what we've just called the “state” so far; it is defined by macroscopically measurable variables such as V , T , and n . The macrostate is stable for a system in equilibrium.
 - A **microstate** is a specific set of submicroscopic particle positions and velocities $\{r_i, v_i\}$. Macrostate is an average over all accessible microstates. The microstate is always changing, even for systems in equilibrium.
- Higher W is associated with all the structural factors we've already identified that contribute to higher S : larger volume, higher temperature, greater mass, more “wiggly” molecules, and more gaseous substances.
- The logarithm of W is proportional to the entropy S . The proportionality constant between the two is **Boltzmann's constant** k_B (1.381×10^{-23} J/K).

$$S = k_B \ln W$$

- Consider a system containing N gas particles (equivalently, n moles times Avogadro's number N_A) occupying a volume V at temperature T . The likelihood W of this state varies exponentially with the number of particles.

$$W \propto V^{n \cdot N_A}$$

$$S \propto k_B \ln V^{n \cdot N_A} = n N_A k_B \ln V = n R \ln V$$

- The gas constant is Boltzmann's constant. Boltzmann's constant is the gas constant.
One is at the scale of 1 mole (R) and the other is at the scale of 1 molecule (k_B). Conceptually, they are the same!
- For an ideal gas undergoing a change in volume from V_1 to V_2 ,

$$\Delta S = n R \ln V_2 - n R \ln V_1 = n R \ln \left(\frac{V_2}{V_1} \right)$$

18.2 The Third Law of Thermodynamics

- The **third law of thermodynamics** establishes an absolute zero of entropy: the entropy of a perfectly crystalline material at zero Kelvin.

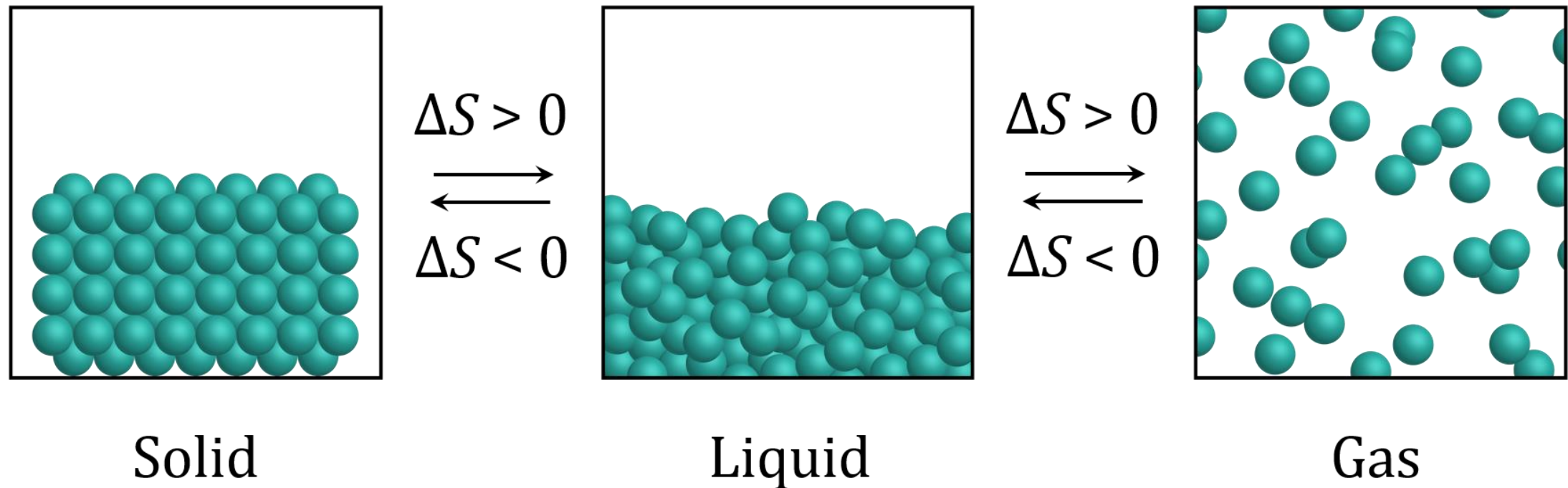


Figure 18.1. Entropy changes for phase transitions involving solid, liquid, and gas.

18.2 Standard Molar Entropy



- **Standard molar entropy (S°)** is an absolute entropy value for 1 mole of a substance at a given temperature based on the zero point established by the third law.
 - Units of Joules per Kelvin per mole
 - Recall that the $^\circ$ symbol indicates that the substance is under standard conditions...

18.2 Factors Affecting Standard Molar Entropy

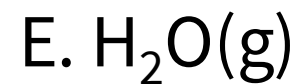
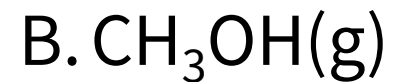
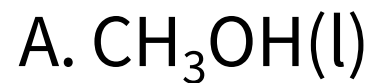
- What factors affect the standard molar entropy of a substance? We already know!

Compound	S° (J K ⁻¹ mol ⁻¹)	Compound	S° (J K ⁻¹ mol ⁻¹)
CO ₂ (g)	213.8	HCl(g)	186.9
CaO(s)	38.1	Hg(l)	75.9
CaCO ₃ (s)	91.7	Hg(g)	175.0
H ₂ (g)	130.7	I ₂ (s)	116.1
H ₂ O(l)	70.0	I ₂ (g)	260.7
H ₂ O(g)	188.8	O ₂ (g)	205.2

Table 18.1. Standard molar entropies for select substances.

- **Standard entropy of reaction (ΔS°)** can be calculated from standard molar entropies. This is just an application of entropy as a state function and Hess's law.

Which substance would you expect to have the *greatest* standard molar entropy?



Chapter 18.B

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18.4 Entropy Changes and Spontaneous Processes



- According to the second law, ΔS_{univ} is positive or zero for any spontaneous process.
- Because $\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr}$, this can happen when **either or both** ΔS_{sys} and ΔS_{surr} is/are positive...

The second law of thermodynamics states that $\Delta S_{univ} > 0$ for all spontaneous processes. Which combination of entropy changes in the system and surroundings *cannot possibly* be associated with a spontaneous process?

A. $\Delta S_{sys} > 0$ $\Delta S_{surr} > 0$

B. $\Delta S_{sys} < 0$ $\Delta S_{surr} < 0$

C. $\Delta S_{sys} > 0$ $\Delta S_{surr} < 0$

D. $\Delta S_{sys} < 0$ $\Delta S_{surr} > 0$

18.4 The Second Law and Entropy



- Exothermic processes ($\Delta H_{sys} < 0$) are more likely than endothermic process ($\Delta H_{sys} > 0$) to be spontaneous...but some endothermic processes are spontaneous.
- Processes with a positive entropy change in the system ($\Delta S_{sys} > 0$) are more likely to be spontaneous than processes with a negative entropy change in the system ($\Delta S_{sys} < 0$) ...but there are some spontaneous processes with $\Delta S_{sys} < 0$.
- To make sense of this, we need to reformulate the second law to (a) incorporate enthalpy and (b) focus on the system. We start with a spontaneous process occurring at constant T and P .

$$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} \geq 0$$

- **Gibbs free energy ($G = H - TS$)** emerges as an important state function from this derivation. Gibbs free energy is the enthalpy of the system minus the product of temperature and entropy. “Free” here means “available.”
 - H is the internal energy U plus PV , the total energy in the system that be transferred into work based on the [first law](#).
 - TS is the quantity of energy in the system that is randomly distributed and thus unavailable to do useful work in accordance with the [second law](#): the “second-law tax.”
 - For a spontaneous process, ΔG is the maximum useful work (w_{max}) that the process can “do” on its surroundings.
 - For a non-spontaneous process, ΔG is the minimum work that must be invested to “make the process go.”
- For a process occurring under standard conditions...

18.4 Standard Free Energies of Formation and Reaction



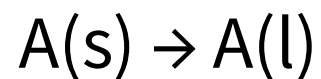
- **Standard free energy of formation (ΔG_f°)** is the free energy change for the formation reaction of a compound occurring under standard conditions.
- Two ways to calculate the standard free energy of reaction (ΔG°_{rxn})...

- The value and sign of ΔG depend on temperature.

ΔH	ΔS	ΔG	Reaction Characteristics
+	+	+ or –	Spontaneous at high temperatures Nonspontaneous at low temperatures
+	–	+	Nonspontaneous at all temperatures
–	+	–	Spontaneous at all temperatures
–	–	– or +	Spontaneous at low temperatures Nonspontaneous at high temperatures

- For processes with ΔH and ΔS having the same sign, the critical temperature T_{crit} at which the process becomes spontaneous is the temperature at which $\Delta G = 0$.

What is the melting point of a compound A? The standard enthalpy and entropy of fusion for A are listed below. Enter your response in Kelvin to the nearest 1 K.



$$\Delta H^\circ = 8.1 \text{ kJ mol}^{-1}$$

$$\Delta S^\circ = 42.4 \text{ J mol}^{-1} \text{ K}^{-1}$$

18.5 Enthalpy- and Entropy-driven Processes

- Processes that are spontaneous only at low temperatures are called **enthalpy driven**. They are “encouraged” by exothermic ΔH .
- Processes that are spontaneous only at high temperatures are called **entropy driven**. They are “encouraged” by positive ΔS .

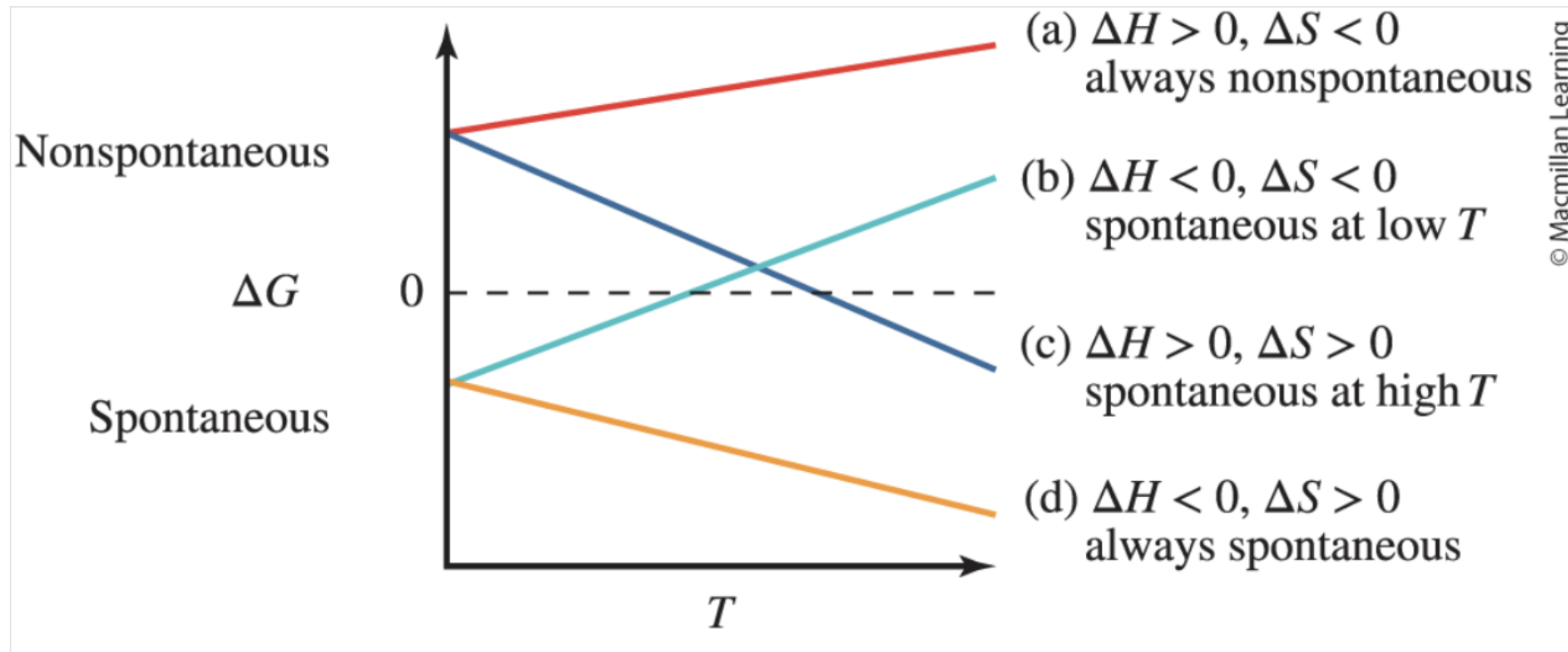
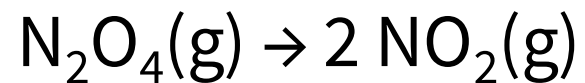


Figure 18.7. Graphical dependence of free energy change on temperature.

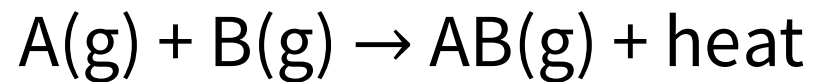
At constant pressure, the reaction below is endothermic.



The reaction (as written) is...

- A. Always spontaneous
- B. Never spontaneous
- C. Spontaneous only above a critical temperature
- D. Spontaneous only below a critical temperature

Which line on the graph of ΔG versus T below corresponds to the hypothetical chemical reaction below?



- A. Line (a) – red
- B. Line (b) – cyan
- C. Line (c) – blue
- D. Line (d) – orange

